

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

THE NORMAL DISTRIBUTION

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Distributional Modeling

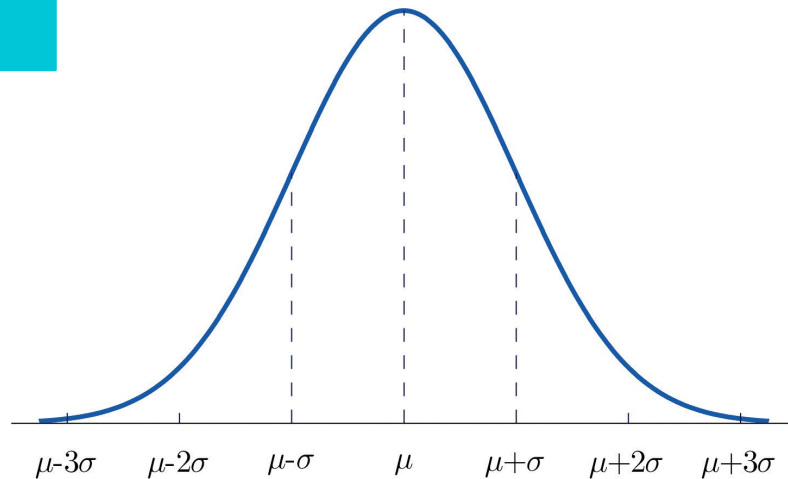
A useful way to understand a data set:

- Fit a probability distribution to it.
- Simple and compact.
- Captures the big picture while smoothing out the wrinkles in the data.
- In subsequent application, use distribution as a proxy for the data.
- Can use to generate new data samples.

Which distributions to use?

- Some of basic distributions: normal, Poisson, binomial, multinomial
- *Products* of simple distributions
- *Mixtures* of simple distributions

The Normal Distribution

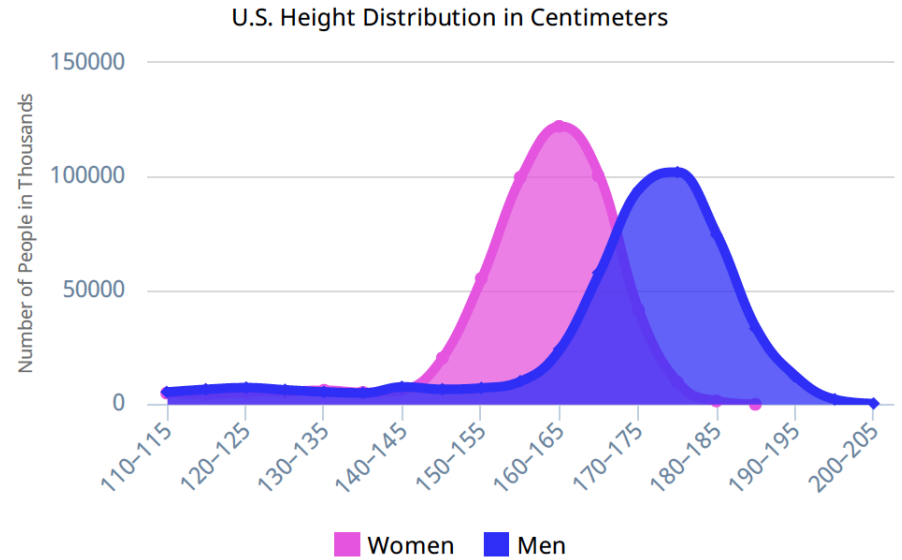


The normal (or *Gaussian*) $N(\mu, \sigma^2)$ has mean μ , variance σ^2 , and density function

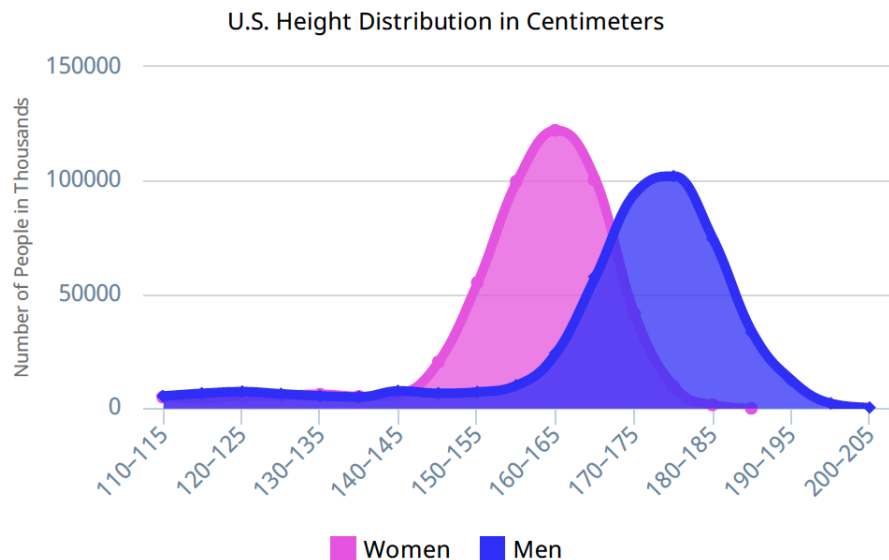
$$p(x) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

- 68.3% of the distribution lies within one standard deviation of the mean, $\mu \pm \sigma$
- 95.5% lies within $\mu \pm 2\sigma$
- 99.7% lies within $\mu \pm 3\sigma$

Gaussians are Everywhere



Gaussians are Everywhere



statpedia.com

Central Limit Theorem: Let $X_1, X_2, \dots$ be independent with $\mathbb{E}X_i = \mu_i$, $\text{var}(X_i) = v_i$.

Then

$$\frac{(X_1 + \dots + X_n) - (\mu_1 + \dots + \mu_n)}{\sqrt{v_1 + \dots + v_n}} \xrightarrow{d} N(0,1)$$

Fitting a Gaussian to Data

Given: Data points $x_1, \dots, x_n$ to which we want to fit a distribution.

What Gaussian distribution $N(\mu, \sigma^2)$ should we choose?